

Health Connect Clinic

Initial Data Analysis

Project Background

HealthConnect Clinic is a fictional healthcare provider that manages appointment-based services. The clinic is experiencing challenges including missed scheduled appointments, difficulty understanding factors associated with no-shows, inefficient use of appointment slots, repetitive appointment related enquiries, and the need to improve patient engagement and administrative support. The overall project explores how data, machine learning and Generative AI can support better decision making and improve patient experience.

Data Analytics Objective

The objective of the Data Analytics track is to understand the HealthConnect appointment data and investigate appointment attendance and no-show patterns. The focus is on understanding the available data, assessing its quality, identifying relevant variables, defining business questions and potential KPIs, and establishing an initial approach for subsequent analysis.

Resources Reviewed

- HealthConnect_Appointment_Data.csv — the fictional and anonymized appointment dataset.
- HealthConnect_Data_Dictionary.xlsx — used to understand variables, data types and descriptions.
- HealthConnect Clinic Knowledge Base — relevant project context.

Dataset Overview

The dataset contains information relating to patient demographics, appointment details, booking information, previous appointment history, previous no-shows, reminder information, distance to the clinic, waiting time and appointment outcomes. The Data Dictionary identifies appointment_id as the unique appointment identifier and primary key, while patient_id may appear across multiple appointments. The principal outcome variable is appointment_outcome, with the documented categories Attended, No-Show and Cancelled.

Data Quality Assessment

Quality Check	Finding	Potential Impact	Proposed Action
distance_to_clinic_km	Approximately 1% empty.	May affect distance analysis.	Retained during initial assessment; did not automatically replace with zero.
waiting_time_minutes	Approximately 1% empty.	May affect waiting-time analysis.	Retained during initial assessment; determine treatment before detailed analysis.

Duplicate appointment IDs	No duplicate appointment IDs identified using appointment_id.	No apparent primary-key duplication issue.	No duplicate removal required based on this check.
Data types	Reviewed against the Data Dictionary.	Incorrect types could affect analysis.	Correct types where necessary and document significant changes.
Categorical consistency	Inspect categorical fields for unexpected/inconsistent labels.	Could distort grouping and calculations.	Standardise confirmed inconsistencies and document changes.

Relevant Variables for No-Show Analysis

Variable	Category	Relevance	Priority
appointment_type	Appointment	May show whether appointment types have different attendance patterns.	High
appointment_date	Appointment	Supports time-based attendance analysis.	Medium
appointment_day	Appointment	Allows comparison of no-shows by day.	High
appointment_time	Appointment	Allows comparison by appointment time.	High
gender	Demographic	May reveal demographic attendance patterns.	Medium
age	Demographic	May reveal differences in attendance across age.	High
age_group	Demographic	Allows comparison across age groups.	High
booking_date	Booking	Provides context for when appointments were booked.	Medium
booking_lead_days	Booking	Can investigate whether longer lead times relate to no-shows.	High
previous_appointments	History	Shows previous appointment history and may identify attendance patterns.	Medium
previous_no_shows	History	Can identify patients with previous missed appointments.	Very High
reminder_sent	Reminder	Can investigate the	Very High

		relationship between reminders and attendance.	
reminder_channel	Reminder	Can compare attendance patterns across reminder channels.	Medium
distance_to_clinic_km	Accessibility	Can investigate whether distance relates to attendance.	High
waiting_time_minutes	Waiting time	May investigate whether estimated waiting time relates to attendance.	Medium
appointment_outcome	Outcome	Determines whether an appointment was attended, missed or cancelled.	Very High

Business Questions

- i. What proportion of scheduled appointments result in no-shows?
- ii. Which patient and appointment characteristics are associated with appointment no-shows?
- iii. Is previous no-show behavior associated with future appointment attendance?
- iv. Is booking lead time associated with appointment no-shows?
- v. Is receiving an appointment reminder associated with appointment attendance?
- vi. Is distance to the clinic associated with appointment attendance?
- vii. Do no-show patterns differ across appointment days, times or appointment types?

Potential KPIs

These KPIs are proposed for future analysis. At Week 4 they are identified and justified rather than calculated or visualized.

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KPI	Measurement Focus	Business Value
No-Show Rate	No-show appointments as a proportion of relevant scheduled appointments.	Core measure of the missed-appointment problem.
Attendance Rate	Attended appointments as a proportion of relevant scheduled appointments.	Overall attendance performance indicator.
Cancellation Rate	Cancelled appointments as a proportion of relevant scheduled appointments.	Shows another key outcome and potential capacity impact.
Reminder-Associated Attendance Rate	Attendance rate compared across reminder	Supports investigation of reminder-related attendance

	status/categories.	patterns.
Repeat No-Show Rate	No-show appointments involving patients with previous no-show history.	Helps identify recurring attendance behaviour.

Initial Analysis Approach

Data preparation: Review data types, missing values, duplicates and categorical consistency while preserving the original dataset.

Descriptive analysis: Examine appointment volumes and appointment outcomes.

No-show pattern analysis: Compare outcomes across demographic, appointment, booking, history, reminder, accessibility and waiting-time variables.

KPI analysis: Calculate and monitor selected KPIs in subsequent stages.

Segmentation: Identify groups or appointment characteristics with relatively higher or lower no-show rates.

Business interpretation: Translate meaningful patterns into practical insights while distinguishing association from causation.

Assumptions

- The appointment outcome field accurately represents final appointment status.
- The Data Dictionary accurately describes the variables.
- The available variables provide a suitable starting point for investigating attendance and no-shows.
- Observed relationships will be treated as associations unless further evidence supports causal conclusions.

Limitations

- distance_to_clinic_km and waiting_time_minutes contain limited missing values.
- The dataset may not capture every factor influencing attendance.
- The dataset is fictional and anonymised.
- Observed relationships do not necessarily establish causation.

Risks and Dependencies

- Poor data quality could lead to inaccurate conclusions.
- Missing information may limit comparisons.
- Inconsistent categories could distort grouping and KPI calculations.
- Analysis depends on correct interpretation of the Data Dictionary and appointment outcomes.
- Future analysis depends on a suitably prepared dataset.

Week 5 Focus

The next stage should move from data understanding into structured exploratory analysis. Priority should be given to establishing the outcome distribution, calculating the selected KPIs, comparing no-show rates across the highest-priority variables and documenting the strongest patterns. Any recommendation should be based on evidence observed in the data.